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28. Industry-Academia Collaboration in Global World: Indian Perspective

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Abstract

Globalisation and modernisation are posing challenges before education for developing global talent. The individuals who are capable of surviving in multicultural society and working in a fast-changing economic scenario can only be able to face the challenge. In this situation educational institutions need to cross its traditional responsibility and move toward working directly for the economic growth and development of the country. For fulfilling this objective Industry and Academia should work in collaboration. In other parts of the world different initiatives have already been taken in this context by India is still in a nascent stage. There is a lot which needs to be done for effective and productive collaboration of Industry and Academia. The best strategy is based on the financial, technical, material and human resources of the country. As both industry and academia are supposed to collaborate so both the partners should zero in on few common objectives and motives. After deciding on that, a beneficial strategy should be planned on the basis of different resources. A collaboration structure with few of the collaboration strategies has been suggested after analysing various successful practices already running in other countries. The strategies if used effectively would be beneficial not only for both the partners but also for the country at large.

Key Words: Industry-Academia Collaboration, Best Practices, Strategies

Introduction

The National Policy on Education of India (1986) characterizes higher education as a 'Crucial Factor for Survival', which provides the Indian people with an 'Opportunity to reflect on the critical, economic, cultural, moral and spiritual issues.' Education has always been meant to fulfil the needs of the society. Since time immemorial it is doing its job efficiently but due to globalization and modernisation education system is facing complex issues and is finding itself in a critical situation. Global society is posing challenges before educational institutions to nurture global talent by fostering technological skills, reflective and creative thinking, economic expertise and human values. The time has come when the complex concepts of economics, science and technology like demonetisation, e-wallets, digital payments, satellites, global warming, digital classrooms, social

networking sites etc have become a part and parcel of common man. These concepts are not confined only to class economists and scientists.

The education system is producing quite a huge number of unskilled, frustrated, depressed and anxious degree holders who, due to lack of experience of real work culture, are only increasing the unemployed population of the country. To reduce the gap between theoretical and practical knowledge government, educationists, policy makers, industrialists, and economists have realised that the collaboration between academia and industry is highly required.

Dr APJ Abdul Kalam in his book titled '**Ignited Minds (Unleashing the power within India)**' rightly says that our intellectual forums, political platforms, academic institutions and Chambers of commerce should all work together to take up the full responsibility of the development of the country.

Thrust of different Policy Reports

The UNESCO report (1997) 'Learning to Be' states as -"The correlation and collaboration between society and education are so complex that simplified explanations cannot possibly give an adequate account of them. This remark is valid for the tasks which education assigns to society and society to education reciprocally and also for statements concerning the objectives to which such tasks are designed to lead". Describing education as an image of society, this report has added that education being a sub-system of society, necessarily reflects the main features of that society. It would be vain to hope for a rational human education in an unjust society.

National Policy on Education adopted by the Parliament in **1986** emphasized the need for the collaboration by stating that-"The curricula of technical and management programmes will be targeted on current as well as the projected needs of the industry or user systems. Active collaboration between technical or management institutions and industry will be promoted in programme planning and implementation, exchange of personnel, training facilities and resources, research and consultancy and other areas of mutual interest... Networking will have to be established between technical education and industry, R and D organisations, programmes of rural and community development, and with other sectors of education with complementary characteristics".

The Technology Policy statement, (DST 1993), has emphasized the collaboration between engineering educational institutions and industry through statements such as:- "Attention will be directed to further enlarge the base of polytechnics, technical and vocational institutes, and engineering institutions and launch programmes for training and retraining industrial and technical

personnel in numbers significantly more than what has been attempted hitherto. Industries will be involved in this process of upgrading the human skills...Pursuit of R and D as a career prospect will be deliberately encouraged through further concrete measures so as to attract scientists and technologists to the challenges of creative science and innovative development with a target of doubling their number in R and D by 2000 AD.”

The DST Policy Statement 1993, further states in the context of Policy Implementation that “RandD institutions including academic institutions would be encouraged to interact with industry and other agencies for contract projects. Income arising out of such initiatives will become additional resources for furthering their infrastructure and other RandD activities... All measures will be taken to accelerate the process of transformation of a laboratory technology to an industrial operation through strengthening and involvement of design, consultancy, and project implementation groups”.

The most of the funding at present for Research and Development in the country is coming from the Government side. There is an urgent need for more contribution from industry in research, besides greater private sector involvement. The industry should engage itself more in terms of not only funding but also in skill development, innovation and entrepreneurship. The gap between the academia and industry has to be bridged to enhance employability of our people. While China spends more than one and a half per cent of its Gross Domestic Product on research and development, India spends less than one per cent, and thus needs a closer partnership between industry and academia (Pallam Raju, 2013).

The review of above mentioned policy reports reflects that collaboration of Industry and Academia is the need of the hour. Resources of both the partners should be used effectively to achieve the objective of nurturing global talent to make them capable of leading the country in global world. For the productive and constructive collaboration it is required that all the policy makers and implementers from all fields of interest like economists, scientists, educationists and government should come together to plan strategies. It is said that a single flower makes no garland so to transform India into a developed country a garland made up of united minds of industrial and academic worlds is required.

Need and Benefits of Industry-Academia Collaboration

Ideally Educational Institutions and Industrial organizations should both work towards the development of individual in particular and society in general. The incongruence in achieving this goal occurs when they both work alone and in certain situations they work in different directions. To make their working styles congruent they should both work in the same direction by keeping common outcomes in mind.

A number of committees have made several recommendations on this issue, significant among which are the **National Knowledge Commission (2008)**, the **Kakodkar Committee Report (2011)** and the **Narayan Murthy Report (2012)**. All these reports emphasize the significance of enhancing industry-academia collaboration for augmenting research, innovation, employability and greater productivity through various measures.

The following reasons clearly mention the need for industry-academia collaboration:

- Due to globalization the industrial world has become more complex.
- Industrialization and modernization has developed new challenging needs.
- Demand of highly skilled, competent and cooperative personnel.
- Huge gap between demand and supply of educated individuals.
- Industry requires experienced and productive human resources which is only possible if they are trained beforehand about the working demands and culture of the industry.
- Need of the fresh recruits equipped with knowledge, skill and attitude.

Bisoux (2003) has explored the relationship between academics and industry. He says that corporations are placing growing emphasis on finding the 'right person'. It forces the business schools to think more carefully on whom they hire and therefore the role of industry in the entire business school model becomes important.

Benefits

Whenever any collaboration is made it needs to be kept in mind that both the parties should be at winning end as no one wants to make an initiative which is at loss. Therefore, both the parties should be made clear that for them collaboration is a win-win initiation. In the following table few of the areas have been chosen to show that how collaboration is useful for both Industry and Academia.

Table No.- 01

Benefits of Industry-Academia Collaboration

Areas of Concern	Industry	Academia
Knowledge Creation	Database of Knowledge would help in using more effective ways of production	Applied knowledge would certainly enhance and strengthen the theoretical concepts
Consultancy	The ideas and views from the experts of concerned area would add new perspectives for better planning and implementation	The first-hand experiences shared by the industry experts would broaden the horizons of creative thinking of the students and faculty
Research	Cost-effective and innovative researches can be done as per requirement. Flexibility in research projects	Facility to get better equipment's for researches. Challenging research projects. Financial support for research projects Publication of research papers. Filing of Patents
Human Resource Development	Recruiting efficient and skilled human resources More experienced, innovative and creative personnel Globally talented and technically modern individuals	Faculty Development Programs Internship Facility for students More exposure of Industry of interest Development of values like collaboration and cooperation Getting a glimpse of work culture before entering into it

Keeping in mind the various issues along with the benefits of collaboration there is a dire need of collaboration strategies in the era of globalization and modernization. If India has to play an important role in global market then it should give priority to collaboration strategies.

Best Practices of Collaboration in different parts of world

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In India the collaboration between industry and academia might be at budding stage but in other parts of world a lot of work has been done in this regard. Many collaboration models are running successfully. In the following table the level and outcome of partnerships between industry and academia can be understood properly. There are few initiatives which have been taken in India also but there is lot more to be done.

Table No. – 02

Outcome of Partnerships Between Industry and Academia

Partnerships	Parties Involved	Outcomes
Caltech Boeing Strategic Agreement 2004-Present	Caltech is a world-renowned educational institution with a long history of fulfilling the needs of industries, governments and academia. Boeing is the world's leading aerospace company, providing products and services to over 150 countries	Caltech faculty played a key role in certification of Boeing's new 787 Dreamliner. Boeing has hired researchers in key areas Students are encouraged internship Patent applications have been filed Boeing provides Scholarships
Scintillator Development for Medical Imaging 2005-Present	University of Tennessee and Scintillation Materials Research Center, Siemens Medical Solutions.	Various researches are being done due to partnership Several patents have been filed Articles are published
Improving Research and Education By	Syracuse University and JPMorgan Chase	The Center of Excellence in Information Security by JPMC Tech Center at SU

Bringing the World of Practice On To Campus 2007-Present		The domain of Global Enterprise Technology (GET) was developed. This domain focuses on large complex information systems that are typically not covered via traditional university courses. All internships provide value to the student (work experience), to the company (GET curriculum enables more productive students), and to the university (enhanced student learning). The Institute for Veterans and Military Families (IVMF) Supporting at-risk Youth through Mentoring Programs etc.
The International Food Safety Training Laboratory: A Partnership that Improves the Safety of Food Globally 2011-Present	University of Maryland's Joint Institute for Food Safety and Applied Nutrition (JIFSAN) and the Waters Corporation (Waters)	Trained professionals from various countries Valuable feedback from trainees has been taken for improvement Students have been hired for key roles
MoU of NASSCOM with UGC and AICTE 2012-Present	National Association of Software and Service Companies (NASSCOM) and the All India Council for Technical Education	Curriculum Review Preparing Training Modules Workforce Development

	(AICTE)	Mentorship Programs Organising Workshops and Seminars
IIM Ahmedabad-IDEA Cellular	Indian Institute of Management Ahmedabad (IIMA) and IDEA Cellular Company	Establishment of Telecom Centre Exploring business issues together
Anna University-TCS	Anna University and Tata Consultancy Services (TCS)	Organizing Workshops for students Faculty development programs Awards for best performances Internship opportunities

The analysis of the above mentioned initiatives clearly indicates that other countries are doing well in this field when it comes to outcomes. They are achieving their goals successfully, whereas in India there are only a handful of cases in which such type of collaboration is done. A thorough and outcome based planning is required. This should be followed by rigorous implementation. Strict supervision and evaluation plan should also be prepared so that funds are utilised properly.

Realities and Expectations

Hon'ble Minister Dr. Pallam Raju pointed out that the current stage of Industry-academia linkages in India today is still quite nascent. He further pointed out that a more sustaining and long term structural plan for continuous development would serve to be mutually beneficial for the industrial sector as well as the academic research driven sector of India. Dr. Pallam Raju, speaking in Workshop on Academia- Industry Collaborations, New Delhi (March 5, 2013) organised by Ministry of Human Resource Development, Government of India, pointed out the following challenges that are faced by the industry and academia today:

- Even though India prides itself on significantly enhancing the service sector, the capacity and competency of the manufacturing sector in India requires serious attention
- Despite excellent work being carried out in academic research labs, there is a considerable lag in converting into a marketable commodity
- There are weak linkages between the research and development taking place in academic laboratories and the various manufacturing companies that could use that research to advance

In the afore-said workshop the ways and means of achieving the goal of collaboration between industry and academia and the need to transform our institutions from centres of knowledge dissemination to hubs of knowledge creation were also discussed. The following areas were stressed upon to achieve this goal:

- Recognizing each other's strengths
- Documenting success stories of industry-academia collaboration for dissemination
- Efficient use of government funds
- Research, Innovation and Entrepreneurship (focusing on Manufacturing, Defence and Sustainable Development and key national goals such as Social Development, Health and Water)
- Skill Building and Employability (focusing on Manufacturing and Services)

While there are many priorities before the MHRD, the three areas requiring top-most attention are:

[a] Improving learning outcomes at all levels;

[b] Meeting the shortage of faculty/ teachers at all levels;

[c] Industry-academia linkages to ensure better employability of our graduates and also skilling of our youth.

It is expected from both the industry and academia that collaboration should be strengthened to make the country an efficient competitor among global leaders.

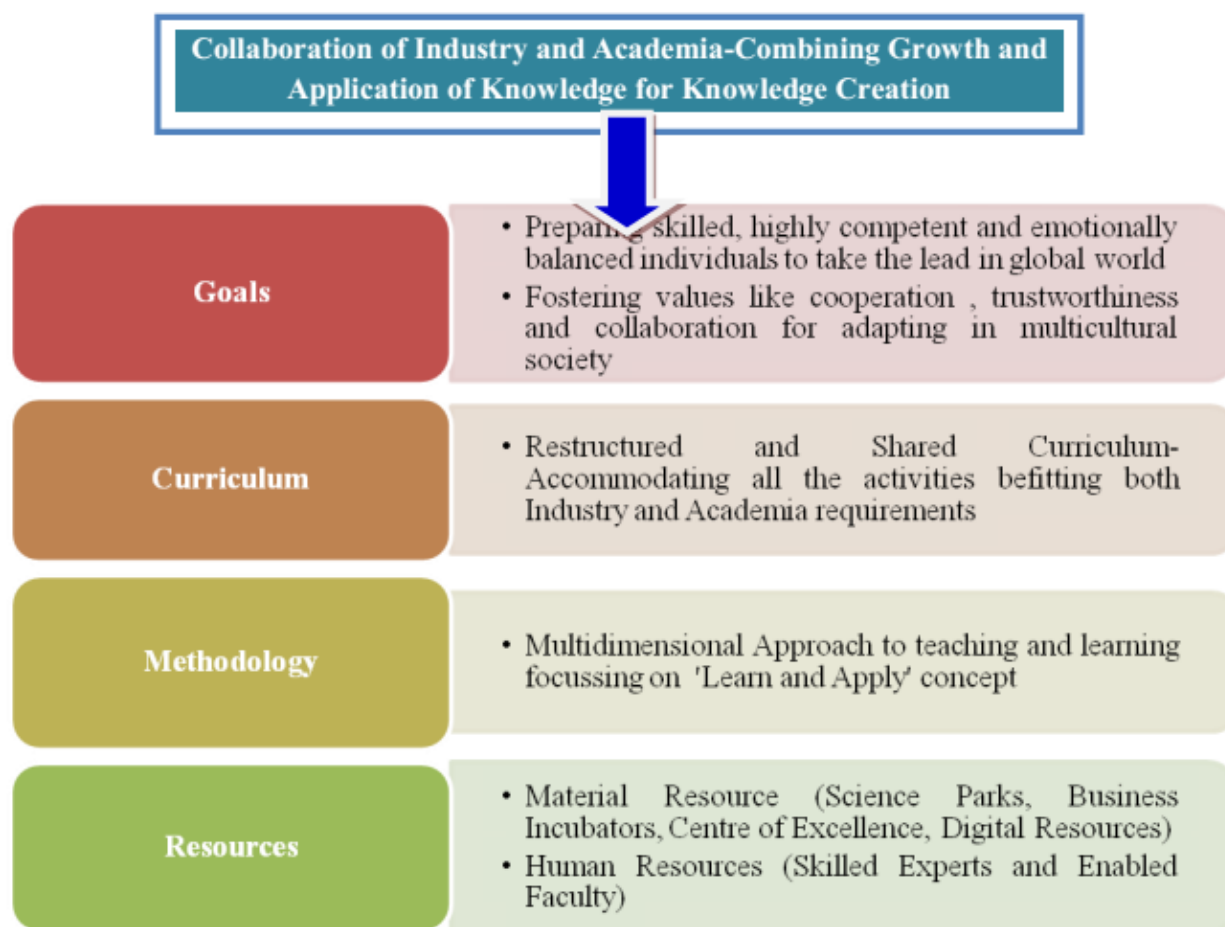
Industry-Academia Collaboration Structure

The following is the analysis of various collaboration models, which are running successfully in other parts of the world and certain points which need to be kept in mind while using them in

Indian scenario. Indian realities are different than other parts of the world. On the basis of realities and expectations a brief outline can be prepared to fulfil the goal of Industry-Academia Collaboration.

Figure No. – 01

Realities and Expectations



Curriculum of academic institutions needs to be restructured so as to enable the industry to integrate its activities. Well-equipped laboratories are required to complete the projects based on industry problems. Both the industry and academia should work in tandem to create an environment of co-learning, co-teaching, co-working and co-researching.

Collaboration Strategies

There are various strategies which can foster the collaboration structure in a better way. If used properly and effectively then it would certainly help in a strong and successful collaboration.

Promote

- Projects that deliver Innovation, Collaboration and Successful Implementation
- Filing of Patents
- Awards to motivate concerned individuals
- Research publications

Figure No. – 01

Collaboration Strategies



Identify

- Problems of Industry for providing Innovative and Creative plans for better production

Facilitate

- Selection and Evaluation of projects that provide sustainable benefits
- Long-term Sustainable projects
- Recruitment of skilled and competent individuals

Enhance

- Experiential Learning and Entrepreneurship Education
- Performance focus

- Global Talents

Create

- Online Portals and other Communication fostering processes
- Conducive Environment for self discipline and self motivation
- Funds from various organizations

Connect

- Research Teams and Industry Teams by allowing access to resources and knowledge database
- Experts from Academia and Industry through Workshops, Seminars, Webinars etc for discussion and sharing stimulating and innovative ideas

Evaluate

- Students should be Evaluated mutually by Academia and Industry

Absorb

- Certificate-holders should be absorbed by the Industry

The discussed strategies would be helpful in making the collaboration effective and productive. The strategies are briefed after reviewing the existing practices of India and other countries. Each pair (Industry and Academia) of partners should make the plan more specific depending upon their common goals, financial, technical, material and human resources to make the collaboration more effective.

Conclusion -Threats and Hopes

Collaboration seems beneficial for Industries and Academic Institutions if planned and implemented properly. On one hand industries are finding that knowledge and research base provided by the academia would help them in getting more production and benefit. On the other hand universities are moving beyond their traditional role of teaching and research toward addressing the demands of industry and contributing directly to economic growth of the country. For successful collaboration both industry and academia need to support each other's missions and motives. Both the partners are looking forward to the effective collaboration as it is beneficial for the socio-economic growth of the country but there are some threatening factors also, which should be kept in mind

before making any collaboration strategy. The research orientations of industry and academia are primarily different, where academia is more interested in developing and creating knowledge base and industry is more focussed on commercial use of that knowledge. It might be possible that academic researches start working in business lines than on fulfilment on academic goals. There is another issue of new patents and products. When it comes to publication, industry is more interested in patenting the research outcomes than on sharing them. There is incongruence in the underlying goals of the industries and academic institutions. The major thrust of academic institutions is on welfare of the society whereas for industries it is on commercial benefits only.

In view of these obstacles to university-industry collaboration, a report of the Joint Project of the U.S. National Council of University Research Administrators and the Industrial Research Institute recommends the following guiding principles for university-industry endeavours (NCURA 2006):

- Successful university-industry collaboration should support the mission of each partner. Any effort in conflict with the mission of either partner will ultimately fail.
- Institutional practices and national resources should focus on fostering appropriate long- term partnerships between universities and industry.
- Universities and industry should focus on the benefits to each party that will result from collaborations by streamlining negotiations to ensure timely conduct of the research and the development of the research findings.

The above mentioned recommendations can be applied to India also keeping in mind the financial and human resources of the country on one axis and capability of producing applicable and useful research outcomes for better production on the other. The ultimate aim of both industry and academia should be the socio-economic development of the country. We can be hopeful for the productive collaboration if we keep the common goal in mind and are able to remove the threats in the larger interest of the country.

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